Mental Health Analysis: BRFSS 2024

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1. Introduction

This report explores **mental health trends** using the **Behavioral Risk Factor Surveillance System (BRFSS) 2024** dataset – one of the largest ongoing health surveys in the United States, conducted by the CDC with over **457,000 respondents** across all 50 states.

Research Questions

- How are poor mental health days distributed across the US population?
- What proportion of respondents have been diagnosed with depression?
- How do mental health outcomes vary by **age**, **sex**, and **education level**?

---

2. Load Libraries

```
library(haven)
library(dplyr)
library(ggplot2)
library(tidyr)
library(scales)
library(forcats)
library(knitr)
```

---

3. Load Data

```
# If brfss is already loaded in environment, skip reloading
if (!exists("brfss")) {
  brfss <- read_xpt("/Users/aadityasingh/Downloads/Data Science Project/LLCP2024.XPT ")
}

cat("Dataset loaded:", nrow(brfss), "rows,", ncol(brfss), "columns")
```

```
## Dataset loaded: 457670 rows, 301 columns
```

---

4. Variable Selection

The following key variables are used in this analysis:

Variable	Description
MENTHLTH	Poor mental health days (past 30 days)
ADDEPEV3	Ever diagnosed with depressive disorder
MISTMNT	Currently receiving mental health treatment
_SEX	Biological sex
_AGEG5YR	Age group (5-year bands)
_EDUCAG	Education level
_INCOMG1	Income group

```
# Check which columns exist in this dataset
wanted <- c("MENTHLTH", "ADDEPEV3", "MISDEPRD", "MISRSTLS",
            "MISTMNT", "_SEX", "_AGEG5YR", "_RACE", "_EDUCAG", "_INCOMG1")

available <- wanted[wanted %in% names(brfss)]
missing   <- wanted[!wanted %in% names(brfss)]

cat("Available:", paste(available, collapse = ", "), "\n")
```

```
## Available: MENTHLTH, ADDEPEV3, _SEX, _AGEG5YR, _RACE, _EDUCAG, _INCOMG1
```

```
cat("Missing:  ", paste(missing,   collapse = ", "), "\n")
```

```
## Missing:   MISDEPRD, MISRSTLS, MISTMNT
```

```r
# Safe select using any_of()
mh <- brfss %>%
  select(any_of(c("MENTHLTH", "ADDEPEV3", "MISDEPRD", "MISRSTLS", "MISTMNT"))) %>%
  bind_cols(
    brfss %>% select(any_of(c(
      SEX       = "_SEX",
      AGE       = "_AGEG5YR",
      RACE      = "_RACE",
      EDUCATION = "_EDUCAG",
      INCOME    = "_INCOMG1"
    )))
  )

cat("\nSubset dimensions:", nrow(mh), "rows,", ncol(mh), "columns")
```

```
##
## Subset dimensions: 457670 rows, 7 columns
```

---

5. Data Cleaning

```r
mh_clean <- mh %>%
  mutate(
    # Recode MENTHLTH: 88 = None (0), 77/99 = NA
    MENTHLTH = case_when(
      MENTHLTH == 88            ~ 0,
      MENTHLTH %in% c(77, 99)   ~ NA_real_,
      MENTHLTH > 30             ~ NA_real_,
      TRUE                      ~ as.numeric(MENTHLTH)
    ),
    # Depression diagnosis
    DEPRESSION = case_when(
      ADDEPEV3 == 1 ~ "Yes",
      ADDEPEV3 == 2 ~ "No",
      TRUE          ~ NA_character_
    ),
    # Sex labels
    SEX_LABEL = case_when(
      SEX == 1 ~ "Male",
      SEX == 2 ~ "Female",
      TRUE     ~ NA_character_
    ),
    # Age group labels
    AGE_LABEL = case_when(
      AGE == 1  ~ "18-24", AGE == 2  ~ "25-29",
      AGE == 3  ~ "30-34", AGE == 4  ~ "35-39",
      AGE == 5  ~ "40-44", AGE == 6  ~ "45-49",
      AGE == 7  ~ "50-54", AGE == 8  ~ "55-59",
      AGE == 9  ~ "60-64", AGE == 10 ~ "65-69",
      AGE == 11 ~ "70-74", AGE == 12 ~ "75-79",
      AGE == 13 ~ "80+",   TRUE      ~ NA_character_
    ),
```

```
    # Education labels
    EDU_LABEL = case_when(
      EDUCATION == 1 ~ "No Diploma",
      EDUCATION == 2 ~ "High School",
      EDUCATION == 3 ~ "Some College",
      EDUCATION == 4 ~ "College Grad",
      TRUE           ~ NA_character_
    ),
    # Treatment (only if column exists)
    TREATMENT = if ("MISTMNT" %in% names(.)) {
      case_when(
        MISTMNT == 1 ~ "Yes",
        MISTMNT == 2 ~ "No",
        TRUE         ~ NA_character_
      )
    } else { NA_character_ }
  ) %>%
  filter(!is.na(MENTHLTH))

cat("Clean dataset:", nrow(mh_clean), "rows after removing missing MENTHLTH values")
```

```
## Clean dataset: 449514 rows after removing missing MENTHLTH values
```

---

6. Summary Statistics

```
cat("Mental Health Days Summary\n")
```

```
## Mental Health Days Summary
```

```
summary(mh_clean$MENTHLTH)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   0.000   0.000   4.415   5.000  30.000
```

```
dep_tbl <- mh_clean %>%
  filter(!is.na(DEPRESSION)) %>%
  count(DEPRESSION) %>%
  mutate(Percentage = paste0(round(n / sum(n) * 100, 1), "%")) %>%
  rename(Diagnosis = DEPRESSION, Count = n)

kable(dep_tbl, caption = "Depression Diagnosis Distribution", align = "lrr")
```

Table 2: Depression Diagnosis Distribution

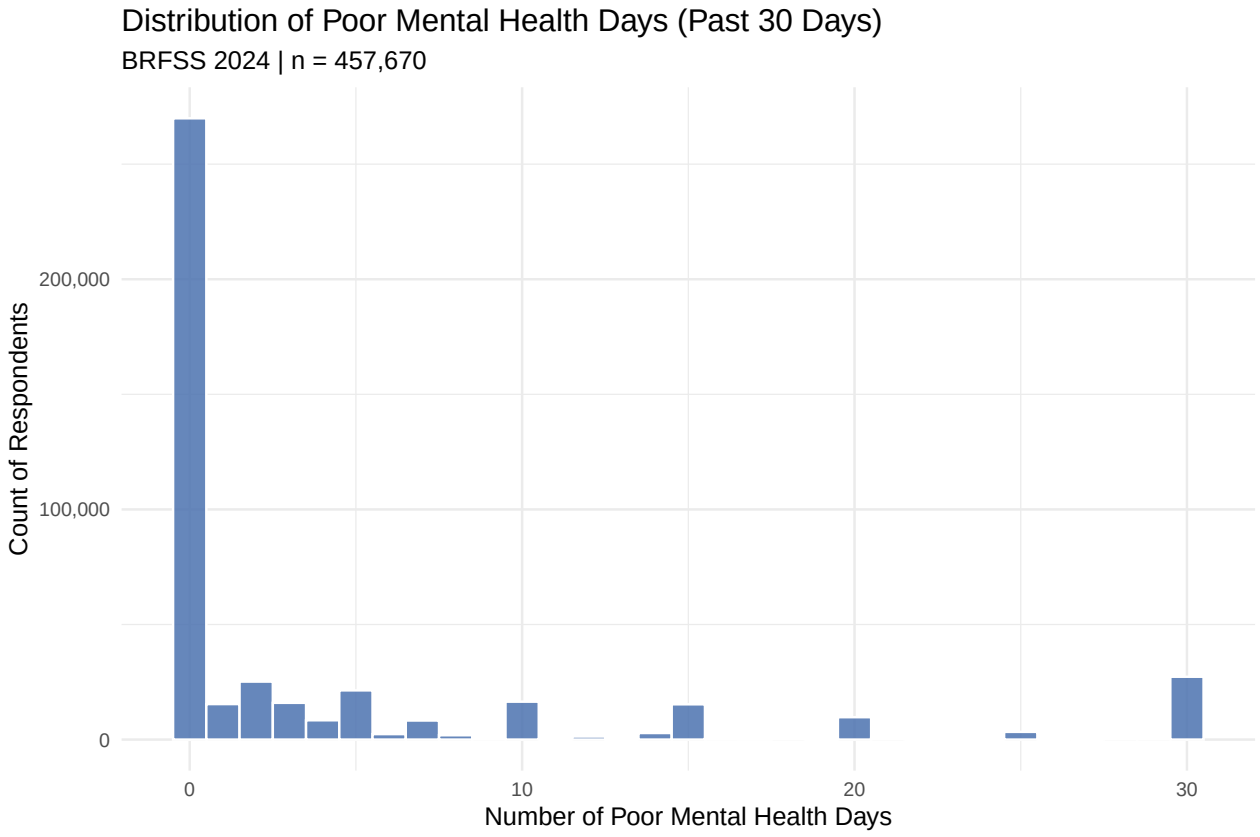
Diagnosis	Count	Percentage
No	353229	79%
Yes	94039	21%

---

7. Visualizations

7.1 Distribution of Poor Mental Health Days

```
ggplot(mh_clean, aes(x = MENTHLTH)) +
  geom_histogram(binwidth = 1, fill = "#4C72B0", color = "white", alpha = 0.85) +
  scale_y_continuous(labels = comma) +
  labs(
    title    = "Distribution of Poor Mental Health Days (Past 30 Days)",
    subtitle = "BRFSS 2024 | n = 457,670",
    x = "Number of Poor Mental Health Days",
    y = "Count of Respondents"
  ) +
  theme_minimal(base_size = 13)
```

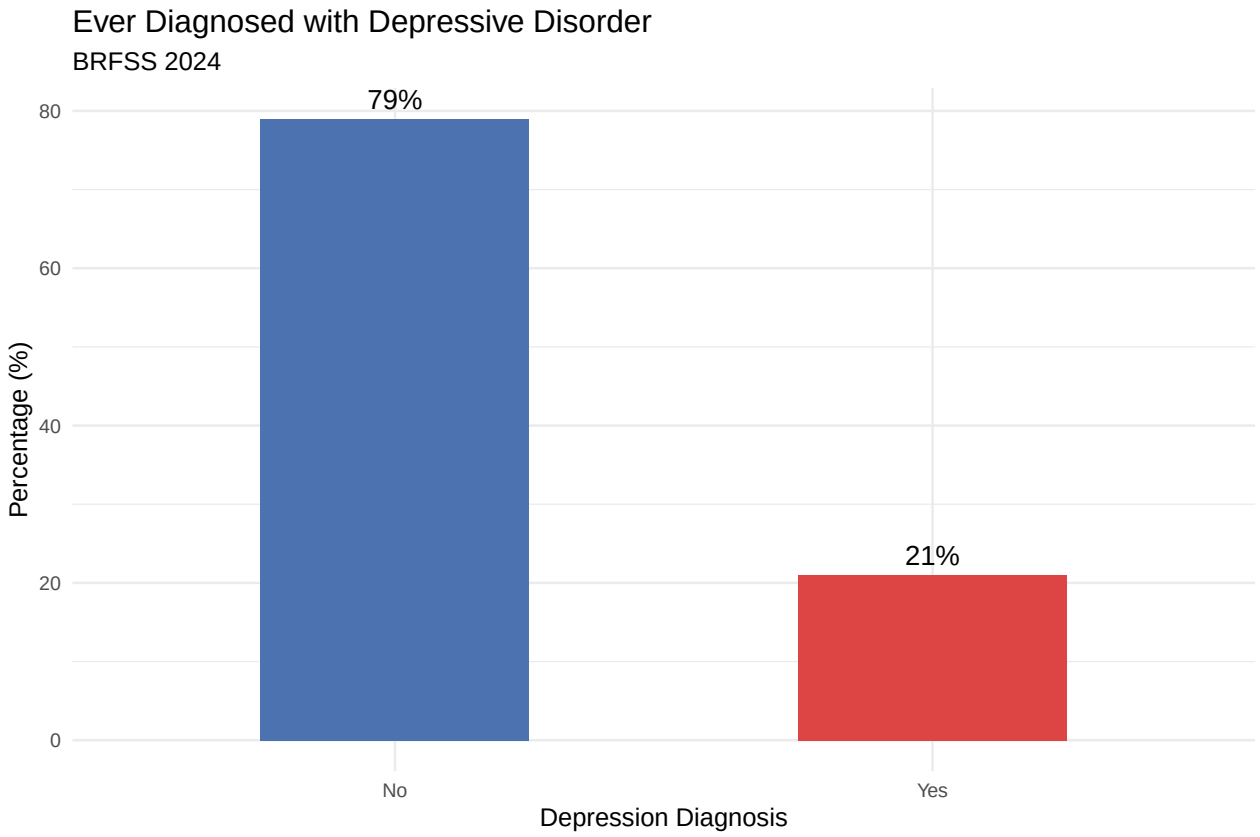


Insight: The distribution is heavily right-skewed, with the majority of respondents reporting zero poor mental health days.

7.2 Depression Diagnosis Breakdown

```
mh_clean %>%
  filter(!is.na(DEPRESSION)) %>%
  count(DEPRESSION) %>%
  mutate(pct = n / sum(n) * 100) %>%
  ggplot(aes(x = DEPRESSION, y = pct, fill = DEPRESSION)) +
  geom_col(width = 0.5, show.legend = FALSE) +
```

```
  geom_text(aes(label = paste0(round(pct, 1), "%")), vjust = -0.5, size = 5) +
  scale_fill_manual(values = c("Yes" = "#DD4444", "No" = "#4C72B0")) +
  labs(
    title    = "Ever Diagnosed with Depressive Disorder",
    subtitle = "BRFSS 2024",
    x = "Depression Diagnosis", y = "Percentage (%)"
  ) +
  theme_minimal(base_size = 13)
```

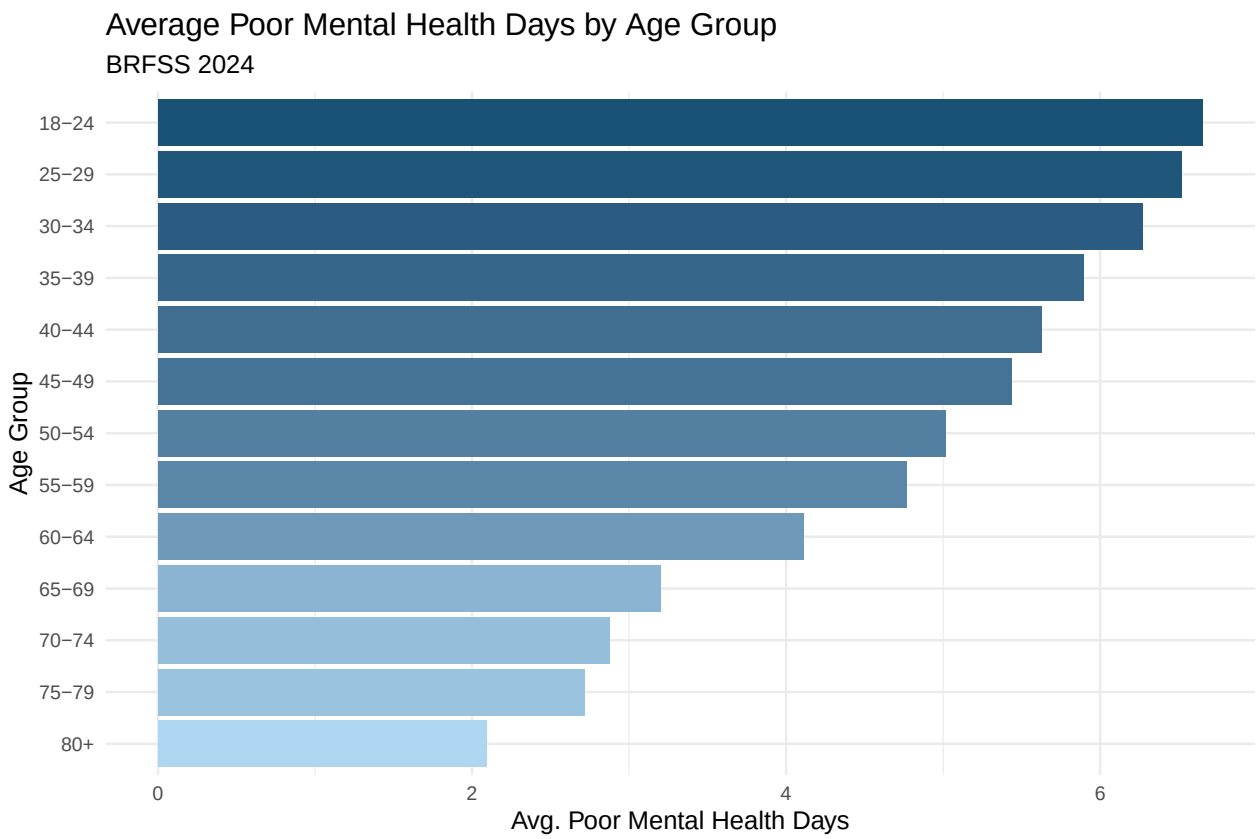


Insight: A significant portion of US adults report having been diagnosed with a depressive disorder at some point in their lives.

7.3 Mental Health Days by Age Group

```
mh_clean %>%
  filter(!is.na(AGE_LABEL)) %>%
  group_by(AGE_LABEL) %>%
  summarise(avg_days = mean(MENTHLTH, na.rm = TRUE), .groups = "drop") %>%
  mutate(AGE_LABEL = fct_reorder(AGE_LABEL, avg_days)) %>%
  ggplot(aes(x = AGE_LABEL, y = avg_days, fill = avg_days)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  scale_fill_gradient(low = "#AED6F1", high = "#1A5276") +
  labs(
    title    = "Average Poor Mental Health Days by Age Group",
```

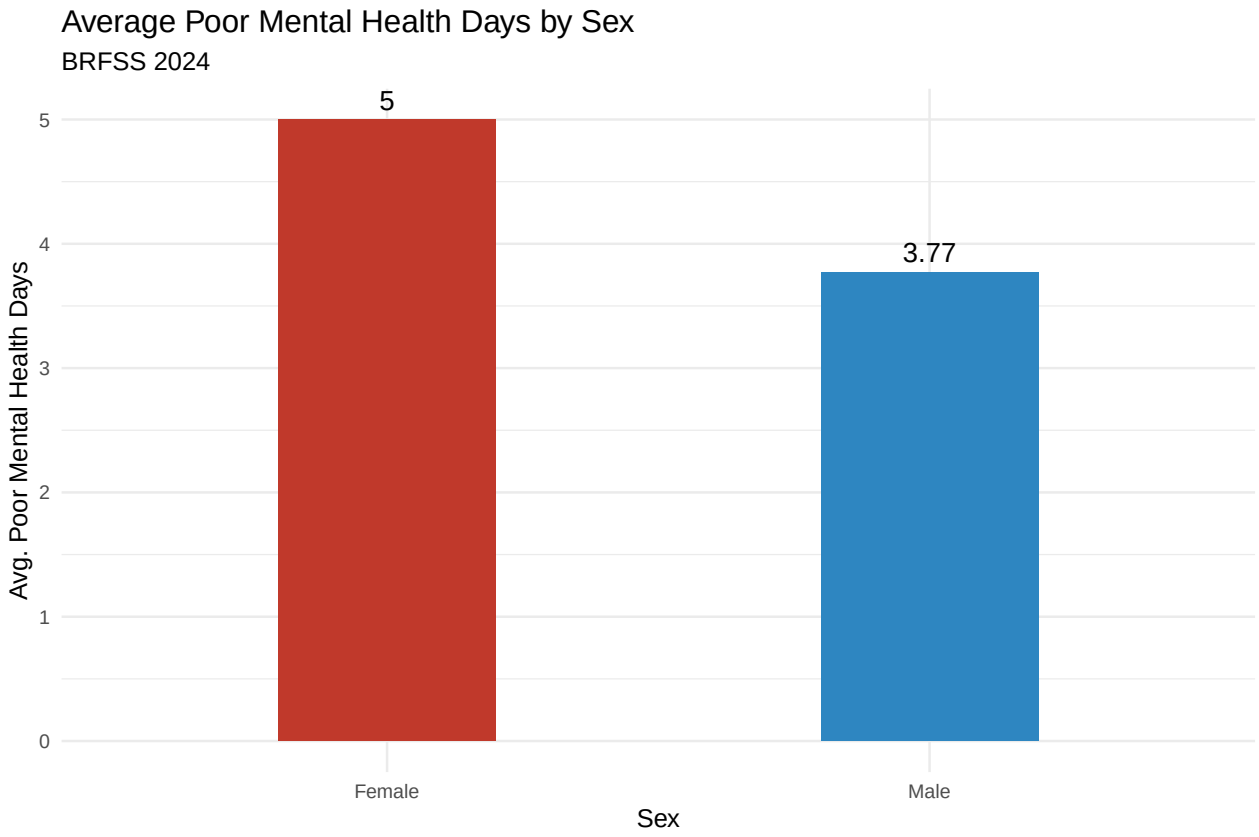
```
    subtitle = "BRFSS 2024",
    x = "Age Group", y = "Avg. Poor Mental Health Days"
  ) +
  theme_minimal(base_size = 13)
```



Insight: Younger adults (18-34) tend to report more poor mental health days compared to older age groups.

7.4 Mental Health Days by Sex

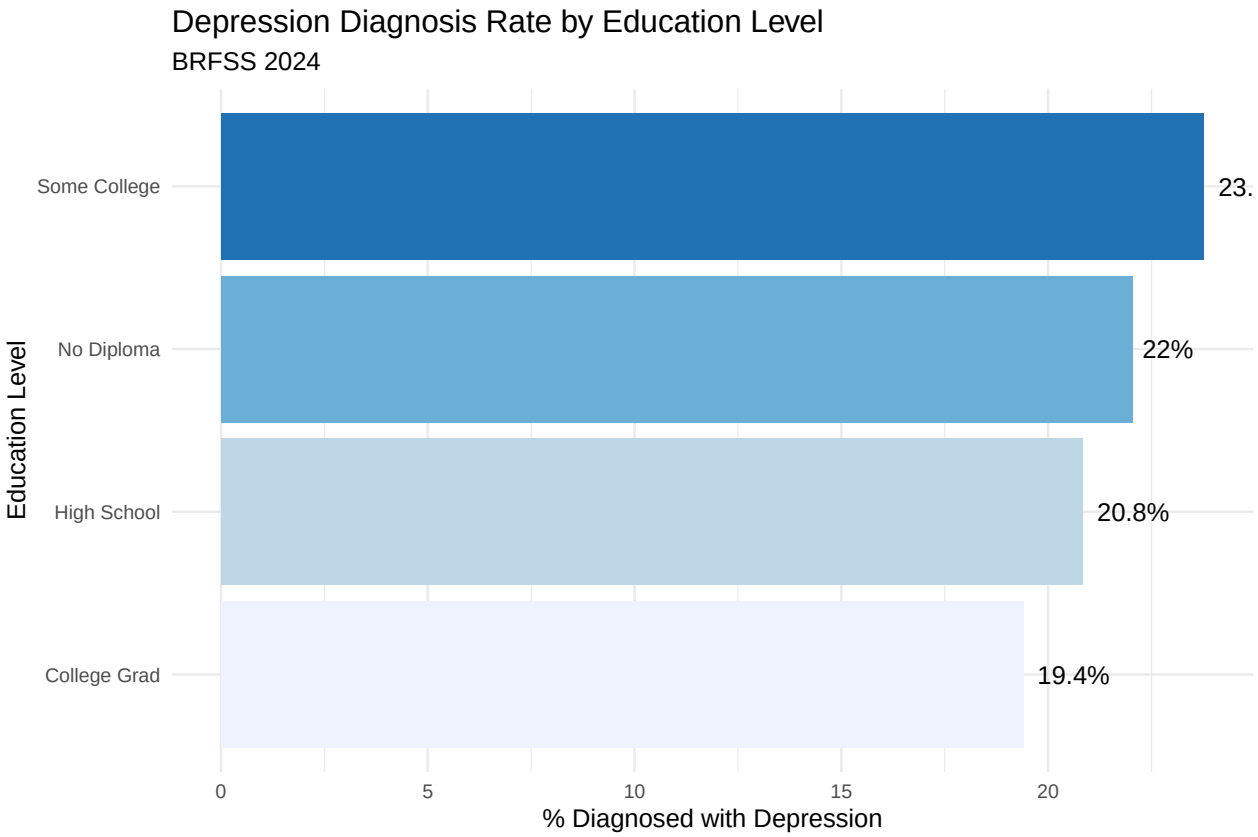
```
mh_clean %>%
  filter(!is.na(SEX_LABEL)) %>%
  group_by(SEX_LABEL) %>%
  summarise(avg_days    = mean(MENTHLTH, na.rm = TRUE),
            median_days = median(MENTHLTH, na.rm = TRUE),
            .groups = "drop") %>%
  ggplot(aes(x = SEX_LABEL, y = avg_days, fill = SEX_LABEL)) +
  geom_col(width = 0.4, show.legend = FALSE) +
  geom_text(aes(label = round(avg_days, 2)), vjust = -0.5, size = 5) +
  scale_fill_manual(values = c("Male" = "#2E86C1", "Female" = "#C0392B")) +
  labs(
    title    = "Average Poor Mental Health Days by Sex",
    subtitle = "BRFSS 2024",
    x = "Sex", y = "Avg. Poor Mental Health Days"
  ) +
  theme_minimal(base_size = 13)
```



Insight: Females report more poor mental health days on average compared to males.

7.5 Depression Rate by Education Level

```
mh_clean %>%
  filter(!is.na(DEPRESSION), !is.na(EDU_LABEL)) %>%
  count(EDU_LABEL, DEPRESSION) %>%
  group_by(EDU_LABEL) %>%
  mutate(pct = n / sum(n) * 100) %>%
  filter(DEPRESSION == "Yes") %>%
  mutate(EDU_LABEL = fct_reorder(EDU_LABEL, pct)) %>%
  ggplot(aes(x = EDU_LABEL, y = pct, fill = EDU_LABEL)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = paste0(round(pct, 1), "%")), hjust = -0.2, size = 4.5) +
  coord_flip() +
  scale_fill_brewer(palette = "Blues") +
  labs(
    title    = "Depression Diagnosis Rate by Education Level",
    subtitle = "BRFSS 2024",
    x = "Education Level", y = "% Diagnosed with Depression"
  ) +
  theme_minimal(base_size = 13)
```

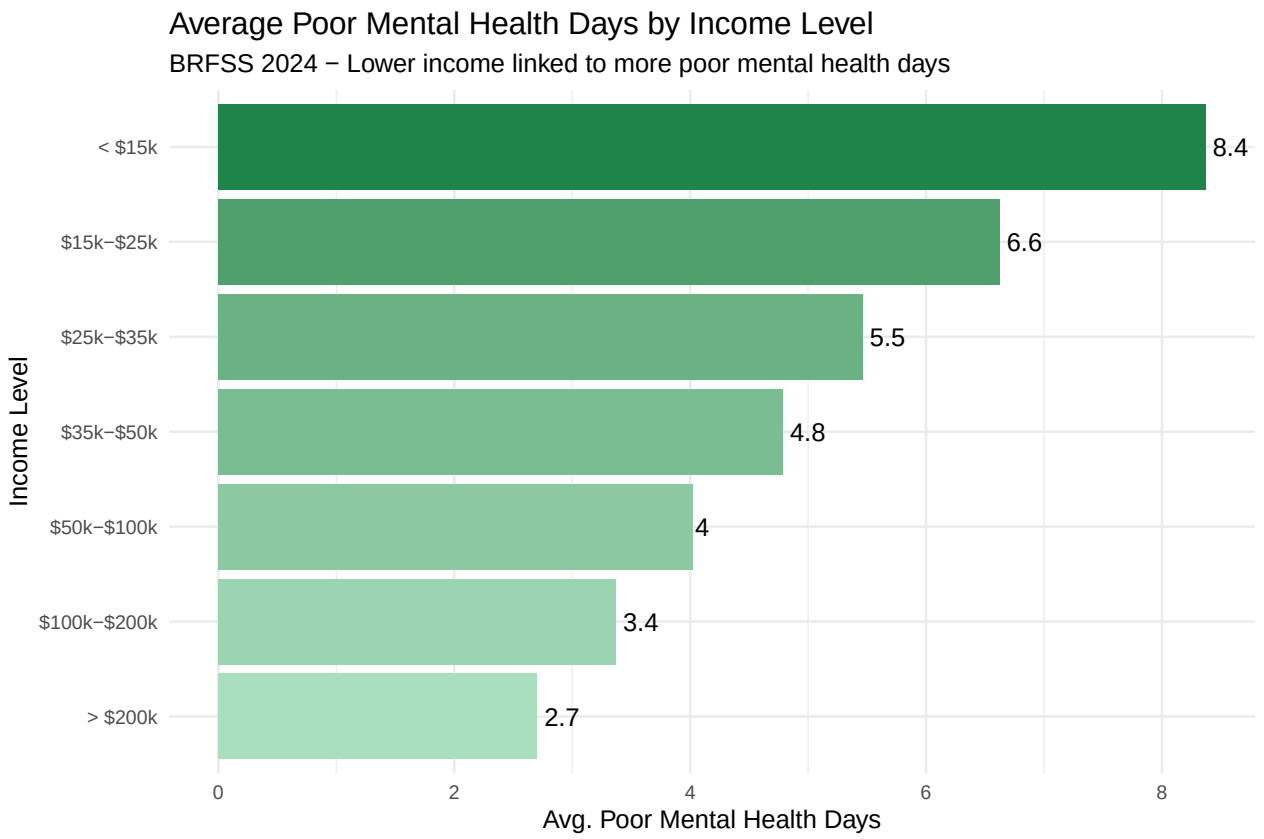


Insight: Depression diagnosis rates vary across education levels, highlighting the link between socioeconomic factors and mental health.

7.6 Mental Health Days by Income Level

```
mh_clean %>%
  filter(!is.na(INCOME)) %>%
  mutate(INCOME_LABEL = case_when(
    INCOME == 1 ~ "< $15k",
    INCOME == 2 ~ "$15k-$25k",
    INCOME == 3 ~ "$25k-$35k",
    INCOME == 4 ~ "$35k-$50k",
    INCOME == 5 ~ "$50k-$100k",
    INCOME == 6 ~ "$100k-$200k",
    INCOME == 7 ~ "> $200k",
    TRUE        ~ NA_character_
  )) %>%
  filter(!is.na(INCOME_LABEL)) %>%
  group_by(INCOME_LABEL) %>%
  summarise(avg_days = mean(MENTHLTH, na.rm = TRUE), .groups = "drop") %>%
  mutate(INCOME_LABEL = fct_reorder(INCOME_LABEL, avg_days)) %>%
  ggplot(aes(x = INCOME_LABEL, y = avg_days, fill = avg_days)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = round(avg_days, 1)), hjust = -0.2, size = 4.5) +
  coord_flip() +
```

```
  scale_fill_gradient(low = "#A9DFBF", high = "#1E8449") +
  labs(
    title    = "Average Poor Mental Health Days by Income Level",
    subtitle = "BRFSS 2024 - Lower income linked to more poor mental health days",
    x = "Income Level", y = "Avg. Poor Mental Health Days"
  ) +
  theme_minimal(base_size = 13)
```



Insight: Lower-income respondents report significantly more poor mental health days, highlighting the strong link between financial stress and mental wellbeing.

---

8. Key Findings

```
avg_days <- round(mean(mh_clean$MENTHLTH, na.rm = TRUE), 2)
pct_zero <- round(mean(mh_clean$MENTHLTH == 0) * 100, 1)
pct_dep  <- round(mean(mh_clean$DEPRESSION == "Yes", na.rm = TRUE) * 100, 1)

findings <- data.frame(
  Finding = c(
    "Average poor mental health days",
    "Respondents reporting 0 bad days",
    "Ever diagnosed with depression",
    "Total respondents analyzed"
  ),
  Value = c(
```

```
    paste(avg_days, "days"),
    paste0(pct_zero, "%"),
    paste0(pct_dep, "%"),
    format(nrow(mh_clean), big.mark = ",")
  )
)

kable(findings, caption = "Summary of Key Findings", align = "lr")
```

Table 3: Summary of Key Findings

Finding	Value
Average poor mental health days	4.41 days
Respondents reporting 0 bad days	60%
Ever diagnosed with depression	21%
Total respondents analyzed	449,514

---

9. Conclusion

This EDA of the **BRFSS 2024** dataset reveals important patterns in mental health across the US population:

- Poor mental health days are **unequally distributed**, with a subset of respondents experiencing significantly more distress.
- **Younger adults and females** are disproportionately affected by poor mental health.
- **Education and income levels** are meaningfully associated with depression diagnosis rates.
- A substantial proportion of US adults have been diagnosed with depression, underlining the scale of the mental health challenge.

Data Source: CDC Behavioral Risk Factor Surveillance System (BRFSS) 2024 Report generated using R 4.5.2